

EXPERIMENT 4

EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

Software Requirements

- Python 3.x
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
-

Theory

Data Inspection is the process of understanding the structure, content, and quality of a dataset before performing advanced analytics.

Statistical analysis helps in understanding:

- Distribution of data
- Central tendency
- Variability
- Relationships among variables

EDA provides meaningful insights that help in decision making and predictive analytics.

Learning Outcomes

After completing this experiment, students will be able to:

- ✓ Inspect datasets
- ✓ Filter records
- ✓ Sort data
- ✓ Group data
- ✓ Calculate statistical measures

✓ Find correlations

✓ Visualize analytical results

Sample Dataset

Student Performance Dataset

Roll No	Name	Department	Marks	Attendance
101	Arun	CSE	85	92
102	Bala	IT	78	88
103	Divya	CSE	95	96
104	Priya	AI&DS	90	94
105	Ravi	IT	72	85
106	Karthik	AI&DS	88	91

Algorithm 1

Dataset Inspection

Step 1

Import dataset

Step 2

Display dataset

Step 3

Check dimensions

Step 4

Display data types

Step 5

Generate summary

Program 1

```
import pandas as pd
```

```
df = pd.read_csv("student.csv")
```

```
print(df.head())
```

Output

Roll No Name Department Marks Attendance

101	Arun	CSE	85	92
102	Bala	IT	78	88
103	Divya	CSE	95	96
104	Priya	AI&DS	90	94
105	Ravi	IT	72	85

Program 2

Shape of Dataset

```
print(df.shape)
```

Output

(6,5)

Meaning:

- 6 Rows
- 5 Columns

Program 3

Data Types

```
print(df.dtypes)
```

Output

Column Type

Roll No	int64
Name	object
Department	object
Marks	int64
Attendance	int64

Algorithm 2

Filtering Data

Step 1

Specify condition

Step 2

Apply filter

Step 3

Display records

Program 4

Students Scoring More Than 85

```
high_marks = df[df["Marks"] > 85]
```

```
print(high_marks)
```

Output

Roll No Name Marks

103	Divya	95
-----	-------	----

104	Priya	90
-----	-------	----

106	Karthik	88
-----	---------	----

Program 5

Students from IT Department

```
it_students = df[df["Department"]=="IT"]
```

```
print(it_students)
```

Output

Roll No Name Department

102	Bala	IT
-----	------	----

105	Ravi	IT
-----	------	----

Algorithm 3

Sorting Data

Step 1

Select column

Step 2

Apply sort_values()

Step 3

Display sorted result

Program 6

Sort Marks in Descending Order

```
sorted_df = df.sort_values(  
    by="Marks",  
    ascending=False  
)
```

```
print(sorted_df)
```

Output

Name Marks

Divya 95

Priya 90

Karthik 88

Arun 85

Bala 78

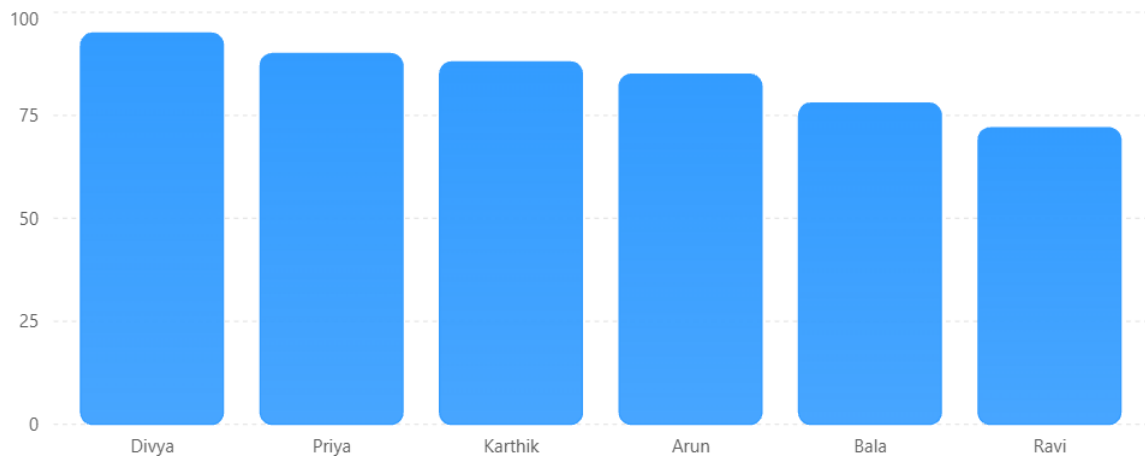
Ravi 72

Visualization

Student Marks Ranking

Student Marks Ranking

Marks obtained by students sorted in descending order.



Algorithm 4

Statistical Analysis

Step 1

Select numerical column

Step 2

Calculate Mean

Step 3

Calculate Median

Step 4

Calculate Mode

Step 5

Calculate Standard Deviation

Program 7

```
print("Mean =",  
      df["Marks"].mean())  
  
print("Median =",  
      df["Marks"].median())  
  
print("Mode =",  
      df["Marks"].mode())  
  
print("Standard Deviation =",  
      df["Marks"].std())
```

Output

Statistic	Value
Mean	84.67
Median	86.5
Mode	No Repetition
Standard Deviation	8.52

Important Formula

Mean

$$\bar{x} = \frac{\sum x}{n}$$

Program 8

Descriptive Statistics

```
print(df.describe())
```

Output

Statistic Marks

Count	6
Mean	84.67
Std	8.52
Min	72
25%	79.75
50%	86.5
75%	89.5
Max	95

Algorithm 5

Grouping and Aggregation

Step 1

Select Department Column

Step 2

Apply groupby()

Step 3

Calculate Average Marks

Step 4

Display Result

Program 9

```
dept_avg = df.groupby(  
    "Department"  
)["Marks"].mean()  
  
print(dept_avg)
```

Output

Department Average Marks

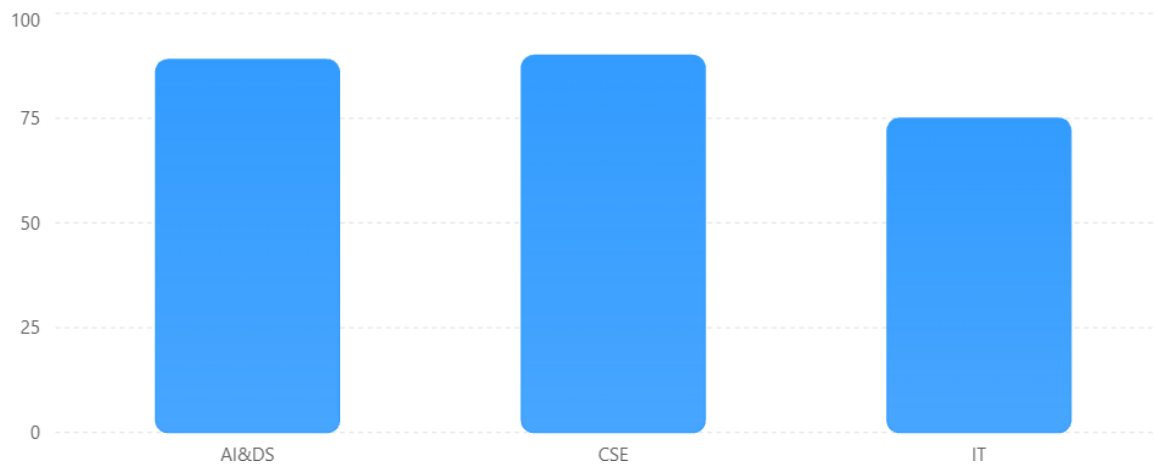
AI&DS	89
CSE	90
IT	75

Visualization

Department-wise Average Marks

Department Average Marks

Average marks obtained by students in each department.



Algorithm 6

Correlation Analysis

Step 1

Select numerical columns

Step 2

Apply corr()

Step 3

Interpret correlation coefficient

Program 10

```
corr = df[[  
    "Marks",  
    "Attendance"  
]].corr()
```

```
print(corr)
```

Output

	Marks	Attendance
Marks	1.00	0.95
Attendance	0.95	1.00

Interpretation

Correlation = 0.95

This indicates:

✓ Strong Positive Correlation

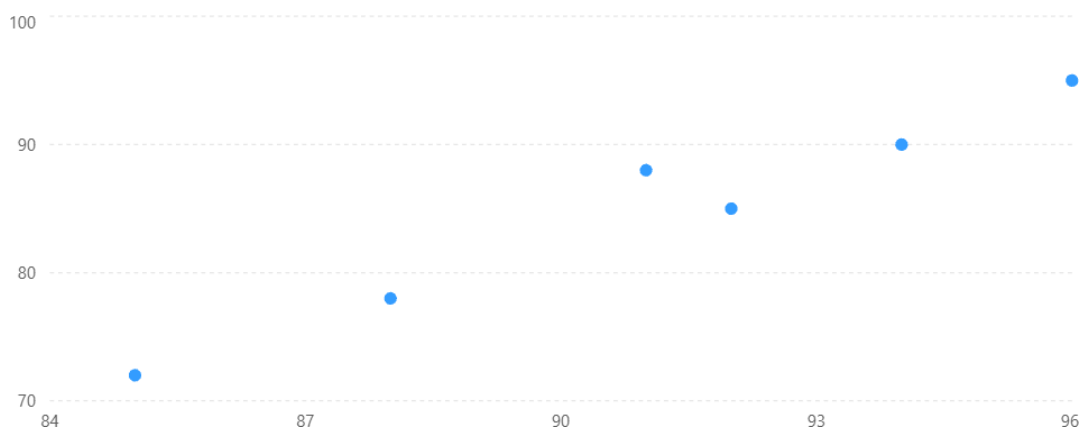
✓ Higher Attendance → Higher Marks

Visualization

Marks vs Attendance Relationship

Marks vs Attendance

Relationship between attendance percentage and marks scored.



Applications

Education

Student Performance Analysis

Healthcare

Patient Health Monitoring

Banking

Customer Transaction Analysis

Retail

Sales Performance Evaluation

Social Media

User Engagement Analysis

Advantages

1. Better Understanding of Data
 2. Improved Decision Making
 3. Trend Identification
 4. Relationship Discovery
 5. Enhanced Predictive Modeling
-

Viva Questions

1. What is EDA?

EDA is the process of analyzing and summarizing datasets.

2. Which function displays dataset dimensions?

shape

3. Which function gives statistical summary?

describe()

4. What is filtering?

Selecting records based on conditions.

5. Which function sorts data?

sort_values()

6. What is grouping?

Combining records with similar values.

7. Which function performs grouping?

groupby()

8. What is correlation?

A measure of relationship between variables.

9. What does correlation value +1 indicate?

Perfect positive relationship.

10. What does correlation value 0 indicate?

No relationship.

11. What is mean?

Average of values.

12. What is standard deviation?

Measure of dispersion from the mean.

13. Why is EDA important?

To understand data before modeling.

14. Which library is used for data analysis?

Pandas.

15. Which library is used for visualization?

Matplotlib.

Result

Thus, data inspection, filtering, sorting, descriptive statistical analysis, grouping, aggregation, and correlation analysis were successfully performed using Pandas. The analytical results were visualized and interpreted successfully.